

JERICO B. BAAL

ELECTRONICS ENGINEER & TECHNICIAN

Design Engineer | Special Science Teacher | Robotics | Embedded Systems



Salawag, Dasmariñas City, Cavite | jerichobaal1730@gmail.com | 09273490769

EDUCATION

Bachelor of Science in Electronics and Communications Engineering | 2015 - 2020

Polytechnic University of the Philippines - Sta. Mesa, Manila

President's List A.Y. 2019-2020

Department of Science and Technology (DOST - SEI) - R.A. 10612 Scholar, 2017 - 2020

LICENCE AND CERTIFICATIONS

- Electronics Engineer 0078418 - Professional Regulatory Commission (2022)
- Electronics Technician 0025290 - Professional Regulatory Commission (2022)
- Data Science Essentials With Python - A Cisco Networking Academy program by DICT (2025)
- Full Stack Web Development Bootcamp - UDEMY , Dr. Angela Yu

PROFESSIONAL EXPERIENCE

Special Science Teacher I - DepEd Bacoor City | October 2022 - Present

- Robotics coordinatorship and mentoring in science-related extracurricular activities.
- Robotics games and innovative research for division, regional, and national competitions.
- Teach various scientific disciplines including Life Science, Chemistry, Physics, and Earth Science,
- Lead and innovate the curriculum planning and class schedule creation for class and teachers program.

Design Engineer - DENSO Philippines | January 2021 - September 2022

- Validated vehicular smart key system designs based on engineering specifications
- Conducted laboratory tests on RF parameters, bandwidth, voltage, and current consumption
- Implemented testing protocols ensuring compliance with regulatory and quality standards
- Used Siemens NX and ELSA schematic tools for engineering drawing and schematic analysis
- Performed benchmarking and monthly technical reporting for new electronic products

TECHNICAL SKILLS

- Embedded Systems & Robotics - Arduino, ESP32, Robotics sensors, modules, IoT development
- Programming - C, Python, HTML, CSS, Javascript
- Electronics and Hardware Design - Schematic & PCB Design (KiCad), Circuit debugging and testing
- Web Development - Node.js, Express.js, React.js, PostgreSQL, Visual Studio, Arduino IDE, Git
- Engineering Tools - Siemens NX , SUMO (Simulation of Urban Mobility)

ACHIEVEMENTS & PROJECTS

- **1st Place, 1kg and 3kg Sumobot RC - National Science Quest 2025** - National level robotics competition by Association of Science Educators in the Philippines, Teacher's Camp, Baguio City
- **Qualifier, Regional Science and Technology Fair 2024 (Region IV-A)** - A Science Investigatory Project under the research and innovation category 'Robotics and Intelligent Machines' - *"PROJECT SAFE: SMS -Based Alerts, Air Quality Monitoring, Fire Surveillance and Environmental Sensing through the utilization of ESP32 Platform"*
- **2nd Place, Division Science and Technology 2025 (Bacoor City)** - Science Investigatory Project under the research category 'Mathematics and Computational Sciences' - *"Comparative SUMO Simulation of Default and "Buhos" Traffic Control Schemes Across Varying Demand Levels at Selected Junctions and Intersections in Bacoor City, Cavite: Traffic Performance Analysis"*

Character reference is available upon request